

BDX Slow Control - Alarm Reference

Operational summary of notifier alarms, thresholds, delays and meaning.

Configuration basis: `notifier/alarms.json` on branch `devel`, updated 2026-10-01.

1. Severity and notification routing

Level	Meaning	Notification
MINOR	Abnormal condition requiring operator attention, but not an immediate safety action.	Telegram
MAJOR	Persistent or serious abnormal condition requiring prompt intervention.	Telegram + email
INTERLOCK	Safety condition requiring or causing a protective action/shutdown.	Telegram + email

Communication and heartbeat policy

All communication-loss, sensor-health and heartbeat conditions use the same operational timing: MINOR after 10 s and MAJOR after 60 s total.

For heartbeat rules: no heartbeat change for 10 s -> MINOR; after another 50 s (60 s total) -> MAJOR. These are state/time based rules: **no percentage threshold is involved.**

Numeric percentage defaults

Generic percentage defaults exist only for numeric deviation rules without explicit overrides. They are **not** a general severity policy. Current defaults: MINOR at 5% for 5 s; direct MAJOR above 10%; sustained MAJOR at 5% for 20 s; recovery 5 s.

2. Global system

Alarm	Trigger / timing	Meaning
Global interlock	INTERLOCK immediately when INTERLOCK_ACTIVE = 1.	A global protective action is active.
System not ready	MINOR after 5 s; MAJOR after 120 s while READY = 0. Suppressed in STANDBY and SAFE.	System is expected to operate but is not ready.
Global heartbeat stale	MINOR at 10 s; MAJOR at 60 s total.	Global IOC is no longer updating.

3. Chiller

Alarm	Trigger / timing	Meaning
Temperature deviation	Numeric deviation from SETPOINT_RBV, only when RUN_RBV is On and DEVIATION_SETTLING is Off.	Measured temperature does not follow the applied setpoint.
Communication unavailable	MINOR 10 s; MAJOR 60 s.	Device communication is unavailable.
Heartbeat stale	MINOR 10 s; MAJOR 60 s total.	Chiller IOC stopped updating.
IOC not running	MAJOR after 5 s if IOC_STATE != RUNNING.	IOC process/state is not operational.
IOC error code	MINOR 10 s; MAJOR 60 s if ERROR_CODE != 0.	IOC/driver reports an error.
Coolant level low	MAJOR after 60 s; INTERLOCK after 600 s.	Low coolant level persisted.
Overtemperature	MAJOR after 60 s.	Hardware reports overtemperature.
General STAT error/alarm/warning	MINOR after 10 s.	Generic chiller diagnostic bit is active.
Reserved STAT bit	MINOR after 10 s.	Reserved status bit unexpectedly active.
External-control value missing	MINOR after 10 s.	Expected external-control value is missing.

Chiller settling suppression

After a setpoint change or RUN -> ON, temperature-deviation alarms are suppressed while settling. Settling ends only after $|T_{\text{measured}} - T_{\text{set}}|$ stays below **0.3 degC for 10 consecutive seconds**. Maximum suppression is **5 minutes**.

4. Low-voltage power supplies (LV1/LV2, channels 1/2)

Alarm	Trigger / timing	Meaning
Device communication	MINOR 10 s; MAJOR 60 s.	Communication with LV1/LV2 is unavailable.
Heartbeat stale	MINOR 10 s; MAJOR 60 s total.	LV IOC stopped updating.
IOC not running	MAJOR after 5 s.	IOC_STATE is not RUNNING.
IOC error code	MINOR 10 s; MAJOR 60 s.	Driver/IOC reports ERROR_CODE != 0.
Voltage deviation	Output On; dV >= 30 mV; >=5% for 10 s -> MINOR; same condition for 60 s -> MAJOR.	Output voltage differs from applied setpoint.
Output mismatch	MAJOR after 5 s if OUTPUT_RBV != OUTPUT_SET, only after first hardware poll following a command.	Hardware output state differs from command.
OVP proximity	Output On; OVP >= 30 mV. >=95% for 5 s -> MINOR; >=100% -> MAJOR.	Voltage approaches/reaches OVP.
Hardware OCP trip	MINOR immediately; MAJOR 10 s; INTERLOCK 20 s.	Hardware OCP protection has tripped.
Current change - 2 s	>=10% within 2 s -> MAJOR, only if dI >= 30 mA.	Abrupt current change.
Current change - 10 s	>=10% within 10 s -> MINOR, only if dI >= 30 mA.	Slower significant current change.
Voltage span - 30 s	>0.1 V -> MINOR; >0.5 V -> MAJOR.	Voltage readback instability.

Instrument-resolution safeguards

Near zero, relative percentages can be misleading. Current-change diagnostics require at least **30 mA** absolute change; voltage deviation requires at least **30 mV**; OVP ratio evaluation is inactive below a **30 mV** OVP reference.

Command transients

Commanded transitions are filtered. LV dynamic diagnostics clear their rolling history after control-state changes. OUTPUT mismatch monitoring is re-armed only after the next hardware poll.

5. Environmental temperature sensors (T00-T03)

Alarm	Trigger / timing	Meaning
Monitoring range	Outside 15-28 degC -> MINOR immediately.	Outside preferred operating band.
Critical range	Below 10 degC or above 35 degC for 5 min -> MAJOR.	Sustained critical temperature.
Change over 10 min	>=3 degC within 10 min -> MAJOR immediately.	Fast temperature drift.
Change over 1 h	>=5 degC within 1 h -> MAJOR immediately.	Large long-term drift.
Sensor unavailable	MINOR 10 s; MAJOR 60 s.	STATUS_OK is false.
Heartbeat stale	MINOR 10 s; MAJOR 60 s total.	Sensor data path stopped updating.
Sensor spread	Spread >4 degC: MINOR after 1 min; MAJOR after 20 min.	Sensors disagree more than expected.

6. Other environmental sensors

Subsystem	Alarm	Trigger / timing	Meaning
Humidity H01	Unavailable	MINOR 10 s; MAJOR 60 s.	STATUS_OK is false.
Humidity H01	Heartbeat stale	MINOR 10 s; MAJOR 60 s total.	Sensor stopped updating.
Pressure P01	Unavailable	MINOR 10 s; MAJOR 60 s.	STATUS_OK is false.
Pressure P01	Heartbeat stale	MINOR 10 s; MAJOR 60 s total.	Sensor stopped updating.

7. Archiver Appliance

Alarm	Trigger / timing	Meaning
Overall health	MINOR 10 s; MAJOR 60 s while ARCHIVER:OK = 0.	Archiver unavailable or degraded.
Health check stale	No LAST_CHECK update for 30 s -> MINOR; MAJOR at 60 s total.	Health monitor itself stopped updating.
Required PV coverage	MINOR after 10 s; MAJOR after 60 s when REQUIRED_HEALTHY < REQUIRED_TOTAL.	One or more required PVs are not healthy.

8. Optional HV subsystem

Alarm	Trigger / timing	Meaning
HV1 communication	MINOR 10 s; MAJOR 60 s.	HV communication unavailable.
HV1 heartbeat stale	MINOR 10 s; MAJOR 60 s total.	HV IOC stopped updating.
HV1 IOC not running	MAJOR after 5 s.	IOC_STATE != RUNNING.
HV1 error code	MINOR 10 s; MAJOR 60 s.	ERROR_CODE != 0.
HV channel voltage deviation	Same 5% / 10 s / 60 s structure with 30 mV absolute floor; optional.	Voltage differs from setpoint.
HV output mismatch	MAJOR after 5 s; optional.	Readback differs from commanded state.
HV OVP proximity	95% for 5 s -> MINOR; >=100% -> MAJOR; optional.	Voltage approaches OVP.

LV-specific OCP-trip, current-change and 30 s voltage-span rules are disabled for HV channels in the current configuration.

9. Optional DAQ subsystem

Alarm	Trigger / timing	Meaning
DAQ communication	MINOR 10 s; MAJOR 60 s.	DAQ communication unavailable.
DAQ heartbeat stale	MINOR 10 s; MAJOR 60 s total.	DAQ IOC stopped updating.
DAQ IOC not running	MAJOR after 5 s.	IOC_STATE != RUNNING.
DAQ error	MINOR after 5 s; MAJOR after 20 s.	DAQ ERROR field is non-empty.
DAQ state mismatch	MINOR after 5 s; MAJOR after 20 s.	STATE_RBV differs from EXPECTED_STATE_RBV.
DAQ not ready during measurement	MAJOR immediately when READY = 0 while MEASUREMENT_ACTIVE = 1.	DAQ not ready during active acquisition.

10. Recovery and grouping

Most alarms require a short recovery interval before a resolution notification is emitted (typically 5 s; some global/archiver rules differ). Alarm groups suppress duplicate notifications when several rules describe the same underlying subsystem condition.

This is a human-readable operational reference. The authoritative machine configuration is `notifier/alarms.json`.